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MR SHOBIT VERMA, DR ARUN DAYAL UDAI

in respect of the application of such design to:

Design and development of 6-DOF Stewart Gough platform for potentiometer-

based Joystick

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1. INTRODUCTION

1.1 Introduction

The Stewart Gough platform is a fascinating mechanical system known for its precision and versatility. With the ability to move in six degrees of freedom (6-DoF)—three translational and three rotational—it plays a crucial role in applications like robotics, flight simulators, and motion control systems. Its robust structure and high accuracy make it ideal for tasks that demand stability and intricate movements.

In this project, Design and Development of 6-DoF Stewart Gough Platform for Potentiometer-Based Joystick, we focused on combining the capabilities of this platform with a potentiometer-based joystick to create an intuitive control system. The joystick translates user input into smooth platform movements, making the system ideal for real-time interaction in areas like virtual reality, haptic feedback devices, and simulators.

1.2 Problem Statement

The primary goal is to design and manufacture a Stewart Gough platform—a parallel manipulator known for its ability to provide stable and accurate spatial positioning. This platform needs to be capable of moving in three translational (x, y, z) and three rotational (roll, pitch, yaw) directions, making it suitable for applications like robotics, flight simulation, and motion control.

To control the platform, a potentiometer-based joystick is used as the input device. The joystick allows the user to intuitively direct the platform's movements, with its signals being translated into precise actuator adjustments. This setup aims to offer a seamless interaction between the user and the platform, enabling real-time and highly sensitive control.

1.3 Objectives (As per this Semester)

1) To finalize the kinematic structure of the design
2) Designing of the CAD model according to available resources at our disposal
3) Arrangement of all the components from online/offline vendors
4) Using Waterjet, Milling and 3D printing to manufacture key components
5) Cutting, welding, grinding, post processing of metal pipes to create the base
6) Error prevention and use of robot arm to mark out drilling points on the base

2. LITERATURE SURVEY

2.1 Background

Stewart Gough platforms are sophisticated parallel manipulators where the end effector is connected to the ground through multiple parallel actuator chains, providing greater rigidity and superior positioning capabilities, as described by Bhaskar Dasgupta et al. [1]. These platforms are widely utilized in applications requiring high load-carrying capacity, excellent dynamic performance, and precise positioning, such as flight simulators and robotic systems. Their inherent design offers significant advantages in terms of stability and strength, making them suitable for demanding tasks.

Among the various configurations of these platforms, the 6 UPS (Universal-Prismatic-Spherical) or 6 SPS (Spherical-Prismatic-Spherical) topology, commonly referred to as the Stewart platform, is particularly popular. As noted by Flabio Dario et al. [2], this configuration is highly efficient in terms of strength-to-weight ratios, making it a preferred choice for applications where structural efficiency is critical [3]. However, the platform does have limitations, such as a relatively small workspace, kinematic singularities, and inconsistent dexterity, which can restrict its flexibility, as highlighted by Yen Ping-Lang et al. [4].

This system operates based on a set of closed-loop kinematic equations, as described by Jing-Shan Zhao et al. [5], which allow precise control of the robot's motion. Additionally, the use of linear potentiometers in the platform's arms, as proposed by Mohammad Behdad Mohammadi Damnab et al. [6], offers an effective way to capture the orientation of the Stewart platform accurately. This innovation transforms the platform into a highly functional joystick for controlling larger parallel robots, addressing the shortcomings of traditional joysticks used for such applications. By leveraging the Stewart platform's precise kinematics and robust control, this setup significantly enhances the accuracy and efficiency of controlling complex robotic systems.

2.2 Kinematics of Stewart Platform

Symbols	Description
a_1, a_2	Length of links 1 and 2
O, O_p	Origin of Base and Moving top platform
B_i	Frame (X_{bi}, Y_{bi}, Z_{bi}) attached to the ith actuator
P_i	Frame attached to the ith joint of the top platform
p_i	Position vector of P_i with respect to O_p
b_i	Position vector of ith actuator frame B_i
q_i	Position vector of actuator frame P_i
t	Position vector of actuator frame O_p
$1, \ldots, 6$	Actuator numbers, i
θ_i	Joint angle of ith actuator
$\mathbf{l}_i, l_i$	Vector joining B_i and P_i, and its magnitude $\|li\|$

Symbols	Description
θ_1	Angle of rotation for the first link about X_{bi} measured in ZY plane
θ_2	Angle subtended by the second link measured in ZY plane
θ_3	Angle subtended by the second link measured in XZ plane
$\bar{P}_i$	$P_i \equiv (x_i, y_i, z_i)$ expressed in frame B_i
$\bar{\mathbf{l}}_i$	$\mathbf{l}_i$ expressed in frame B_i
$P_i'(0, y_i, z_i)$	Projection of $\bar{P}_i$ on ZY plane
a_2'	Projected length of a_2 on ZY plane

Table 2.1 Symbolic Notations

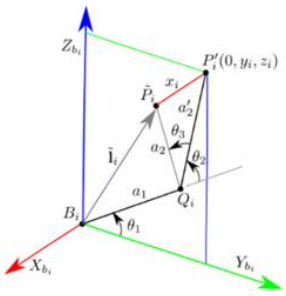


Fig 2.1 Kinematic diagram of an isolated 3-R manipulator.

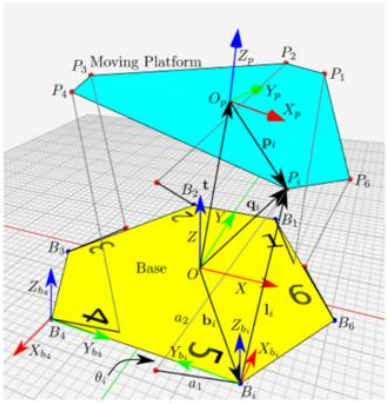


Fig 2.2. Kinematic diagram with symbols used for 6-RSS Stewart

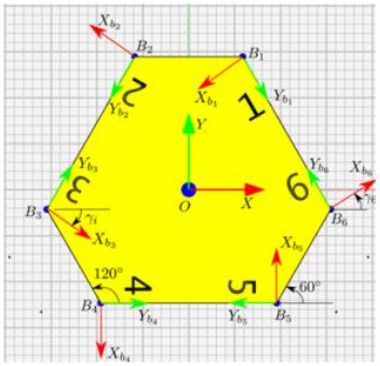


Fig 2.3. Axis frames for actuators on Stewart base

Following are the symbols used to represent different components of the kinematic structure of 6-RSS Stewart platform as shown in Figure 2. It shows a representative shape of the top platform using a convex polygon formed by the joint locations. The precise locations of the actuator frames $\mathbf{B_i}$ and top platform frames $\mathbf{P_i}$ were obtained through CAD software and later through external measurements using Coordinate Measuring Machine (CMM). The position vector of the $\mathbf{i^{th}}$ joint on the top platform with respect to the $\mathbf{i^{th}}$ actuator frame is given by –

$$\mathbf{l}_i = \mathbf{t} + {}^{P}\mathbf{R}_O \cdot \mathbf{p}_i - \mathbf{b}_i$$

where ${}^{\mathbf{P}}\mathbf{R_O}$ is the combined rotation matrix of the top frame with respect to the Stewart base frame O(X,Y,Z).

$$^{P}\mathbf{R}_{O} = \mathbf{R}_{z}(\phi) \cdot \mathbf{R}_{y}(\theta) \cdot \mathbf{R}_{x}(\psi)$$

where,

$$^{P}\mathbf{R}_{O} = \begin{bmatrix} \cos\phi\cos\theta & -\sin\phi\cos\psi + \cos\phi\sin\theta\sin\psi & \sin\phi\sin\psi + \cos\phi\sin\theta\cos\psi \\ \sin\phi\cos\theta & \cos\phi\cos\psi + \sin\phi\sin\theta\sin\psi & -\cos\phi\sin\psi + \sin\phi\sin\theta\cos\psi \\ -\sin\theta & \cos\theta\sin\psi & \cos\theta\cos\psi \end{bmatrix}$$

Rz(ϕ)Rz(ϕ) is the rotation matrix corresponding to the yaw angle ϕ about Z axis, similarly Ry(θ)Ry(θ) and Rx(ψ)Rx(ψ) are rotation matrices corresponding to the roll and pitch angles θ and ψ respectively. The vector $\mathbf{l_i}$ was expressed in actuator frame $\mathbf{B_i}$ to obtain the joint angle of the $\mathbf{i^{th}}$ actuator using.

$$\tilde{\mathbf{l}}_{i} = [\mathbf{l}_{i}]_{B_i} = \mathbf{R}_{z}(\gamma_i) \cdot \mathbf{l}_{i}$$

Knowing the pose of the top platform and corresponding locations of $\mathbf{P_i}$, the problem of inverse kinematics reduces to finding the solution of remaining 3 DoF arm with one active and two passive joints. The rotation of second link about its own longitudinal axis do not affect the arm kinematics and is redundant. Figure 1 shows the kinematic structure of each leg of the Stewart platform comprising of two links connected with one active revolute joint driven by the actuator and a passive spherical joint at the end of first link. Once $\mathbf{l_i \equiv [x_i, y_i, z_i]}$ is evaluated. Algebraic solution for joint angles were performed. From Figure 1, $\mathbf{x_i = a_2 \sin\theta_3}$. Which gives,

$$\theta_3 = \sin^{-1}\frac{x_i}{a_2} \text{ and } a_2' = a_2\cos\theta_3$$

Also,

$$y_i = a_1\cos\theta_1 + a_2'\cos(\theta_1 + \theta_2)$$
$$z_i = a_1\sin\theta_1 + a_2'\sin(\theta_1 + \theta_2)$$

Squaring and adding y_i and z_i gives –

$$\theta_2 = \cos^{-1}\left[\frac{y_i^2 + z_i^2 - (a_1^2 + a_2'^2)}{2a_1a_2'}\right]$$

Expanding y_i gives,

$$y_i = (a_1 + a_2'\cos\theta_2)\cos\theta_1 - a_2'\sin\theta_2\sin\theta_1$$

Assuming, $(a_1 + a'_2\cos\theta_2) = A\cos\varphi$ $(a_1 + a_2'\cos\theta_2) = A\cos\varphi$ and $a'_2\sin\theta_2 = A\sin\varphi$, which on squaring and adding leads to:

$$A = \sqrt{(a_1 + a_2'\cos\theta_2)^2 + (a_2'\sin\theta_2)^2}$$

$$\varphi = \tan^{-1}\left(\frac{a_2'\sin\theta_2}{a_1 + a_2'\cos\theta_2}\right)$$

y_i can now be written as:

$$
\begin{aligned}
y_i &= A[\cos\varphi\cos\theta_1 - \sin\varphi\sin\theta_1] \\
&= A\cos(\theta_1 + \varphi)
\end{aligned}
$$

Which reduces to,

$$\theta_1 = \cos^{-1}\left(\frac{y_i}{A} - \varphi\right)$$

Thus, the final solution for joint angle of the **ith** actuator was obtained. On successive evaluations of the top frame locations **P$_i$** for any given pose of the moving platform and base frame locations **B$_i$**, the joint angles of all the active joint actuators for **i** = 1, 2, 3, 4, 5, and 6 were obtained.

2.3 Significance of the Work

Using potentiometer arms on a Stewart platform as a joystick has great potential for future development. It can be improved with better sensors, haptic feedback, and smarter controls to make it more accurate and user-friendly. This system could be used in fields like robotics, heavy machinery, and aerospace, offering precise and easy control of large-scale Stewart platforms. Traditional joysticks are not very efficient and accurate for the users to give the orientation and pose as input to another large parallel robot, thus this product will assist in the same.

3. MODELLING OF THE STEWART PLATFORM

3.1 Introduction

The first step in building the robot is to make a virtual model of it in CAD software. These CAD models help us to further use various technologies to turn our ideas into reality. We can extract the files made into various formats such as. step, .stl, .dxf, etc. in order to facilitate the input for various machining processes. For the modelling of our Stewart platform, we use SolidWorks 2023 as a CAD software. We first need to realize the appropriate dimensions of the robot.

3.2 CAD Modelling

To finalize the CAD model, we first look at various aspects of the stewart platform and how it affects the overall working of the system. We first layout a 3D kinematic structure and iterate in order to get the maximum possible range of maneuverability. We try to design the model such that we can manufacture it easily with the resources available at out disposal. We also need to consider

the materials and its cost when it comes designing these robots. Overall, after multiple iterations we were content with the design of our system as shown in Figure 3.1

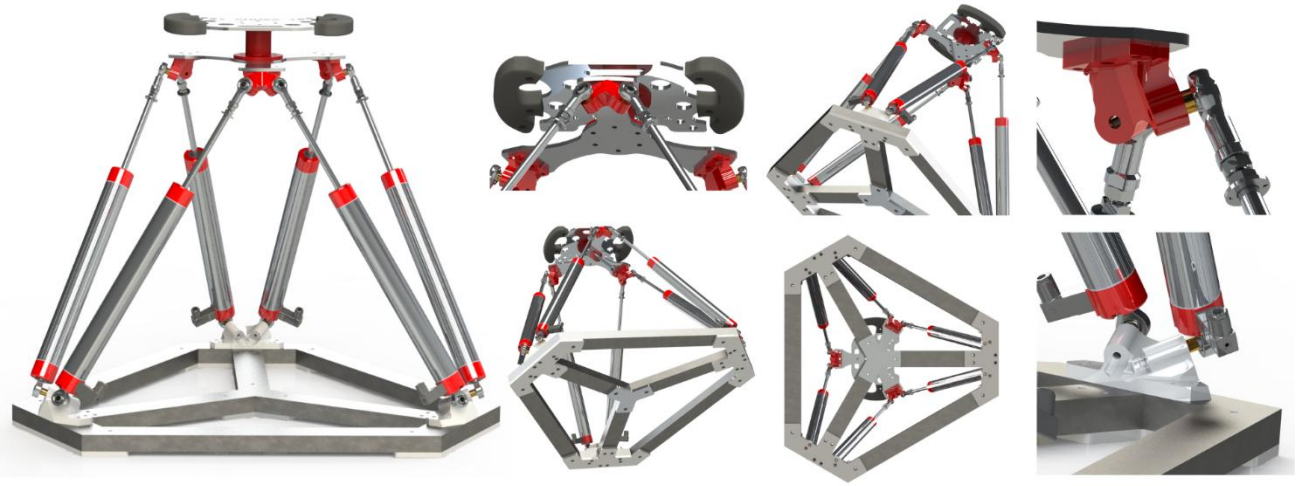


Figure 3.1 CAD model of the 6-DOF Stewart Platform

The model includes 6 potentiometers mated to 6 wedges, by use of bushings. The wedges are inclined in 3 axes and are used in-order to maximize the workspace while reducing significant singularities. The top 3 wedges are fixed on a base frame which can be modelled using MS pipes. We also have 3 supporters below the base which act as a counter bore to our bolts that are going to be inserted. On top 3 wedges are joint to the top plate which then extended to form the joystick handle which the user will hold.

3.21 Kinematic Structure

To model the CAD we require an elaborate plan to structure all the arms of the robot together in order to give space to each component properly. We need to take caution so that the actuator ends do not collide leave ample space to make the wedges and not over exceeding it. The wedges are crucial to make the design more maneuverable. The kinematic structure is shown in Fig. 3.2.

As seen in the figure 3.2, The actual design is truncated from the original kinematic polygon. This is done in order to provide good space to each component. We also need to take care that varying the inclination angle also affects the structure of the robot. In our case the key dimensions are -

Base diameter 777 mm, length 620 mm, Top Diameter 293mm, Inclination 60 degree. These were optimized after various trial and error to correctly place the actuators. Increasing the inclination angle also increases the base diameter. Having larger base diameter: top diameter ratio signifies larger workspace, but reducing the payload capacity. In out case, since the Stewart platform is going to be used by a human, we need not worry about the payload thus our model is designed accordingly.

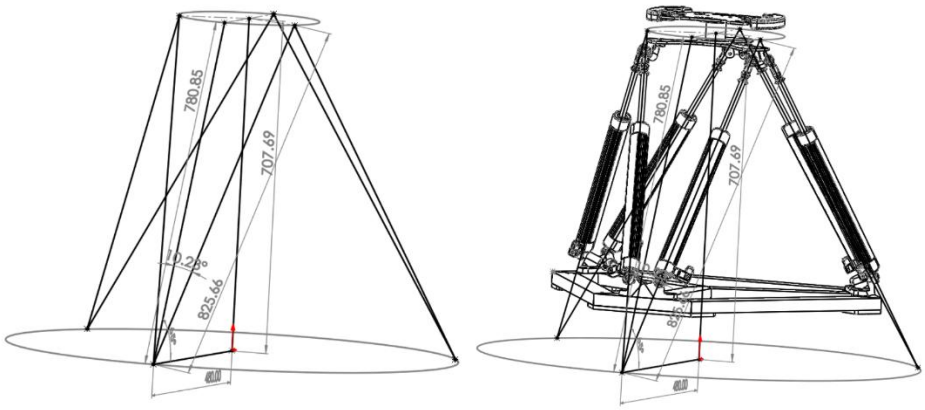


Figure 3.2 Kinematic Structure of our Stewart Platform

In Figure 3.2, we can also observe that the handle of the robot exceeds the kinematic top circle, thus we need interpolate the end orientation to the orientation of the potentiometer ends. The handle is designed such a way that it fits in hand easily. Moreover, at the home position all the potentiometer lengths should be almost halfway open in order to maximize workspace. Overall, the model is quite apt for our application and encompasses every detail.

3.22 Critical Areas in design

Firstly, we need to take care of that the actuator are correctly spaced in order to have the right amount of space for the wedges as explained in section 3.21. The role of wedge is key. These wedges are inclined in 3 axes in order to give maximum output of workspace. They are kept perpendicular to the plane of the inclined actuators, at home position i.e. at midway extension. We are also using bushings as spacers in order to increase the revolving range of the special joint on actuators and avoid collision with the wedge before maximum extension. Another challenging aspect is the base which is to be designed at cut in such a way that we get minimum error after

welding. The base is also attached with supports made of nylon, which act as counter bore and provide sufficient friction in order to avoid slipping of the structure.

3.3 List of components with specifications

The list and details of all the components can be summarized with the table 3.1 given below. This list is designed in such way that we can easily procure the components here in Dhanbad.

Part	Quantity	Material and dimenions	Remarks
Aluminum plate 	1	Aluminum plate 5mm thickness 300mm X 250mm	
Aluminum plate 	1	Aluminum plate 5mm thickness 300mm X 300mm	Solid aluminum, non scratch
Stainless Steel plate 	3	Stainless Steel plate 8mm thickness 320mm X 150mm	
Stainless Steel plate 	1	Stainless Steel plate 8mm thickness 160mm X 160mm	
Rectangular pipe 	4 or 5 m	Rectangular pipe 4m or 5m 2inch x 1 inch Cross Section	CS can be 2in x 2in
Red/White PLA 	1 roll or 1 roll each	PLA	Can try different colours
TPU/TPE 	1 roll (net quantity 250gms)	TPU/TPE	Black looks good?

Bushings 	either pipe - 1m ***or*** 6+2 bushing 5mm long & 6+2 bushing 9mm long	Steel/Brass/bronze CS => 10.62 mm OD, 8.2 mm ID	(6+2 bushing means 6 required + 2 spare) Smallest ID = 8mm Largest OD = 11 or 12 or 13mm
M8 Shoulder Bolt 	6+2 pc with Shoulder length = 50mm 6+2 pc with Shoulder length = 30mm	Stainless Steel M8 X 50mm - 8pcs M8 X 30mm - 8pcs	(Length mentioned is Shoulder length)
M6 Bolts	17	M6 x 45mm	
	10	M6 x 40mm	
	17	M6 x 35mm	
	10	M6 x 20mm	
	26	M6 x 15mm	
M5 Bolts	17	M5 x 15mm	
M6 nuts	90		
M5 nuts	17		

Table 3.1 Component list with specifications

4. MANUFACTURING OF THE ROBOT

4.1 Introduction

After the CAD design is finalized, we proceed towards finding ways to optimally manufacture the components with the machinery available at our central workshop. We need to take care about the various parameters of the design and consider the limitations of the manufacturing machines both in terms of time and accuracy. Cost can also play a major factor while procuring materials. We finalized on using 5mm thick aluminum plate both for top plate and the joystick handle. For the wedges, we initially planned on milling, but due to concern in design might require update, we preferred to used 3D printing initially. Nonetheless, 3D printing is also a good alternative. We used white PLA for the wedges and soft TPE for the rubber handles of the joystick. For the base frame, we decided to go with cutting mild steel pipes of cross section 1 inch x 2 inch and weld

them together to form the frame. We need to take into consideration about the error that occur in frame after cutting and welding as well so as to avoid inaccuracy in drilling.

4.2 Technologies used in manufacturing

We used various technologies such as waterjet, 3D printing, milling, metal cutting, welding, grinding, etc, some as shown in Figure 4.1, in order to finalize the model. For the Top plate and Joystick handle we used two 5mm thick Aluminum sheets. The dimension of the Top plate is 293mm x 293mm and for the handle it is 280mm x 230mm. So, we decided on getting two 300x300mm Al sheets. The models were uploaded for cutting in .dxf format. The waterjet machine also got clogged with abrasive in the middle of manufacturing (common phenomenon), so I also got a chance to open and see the components inside while cleaning them. The final parts came out with high accuracy and rough edges, so required a bit of filing.

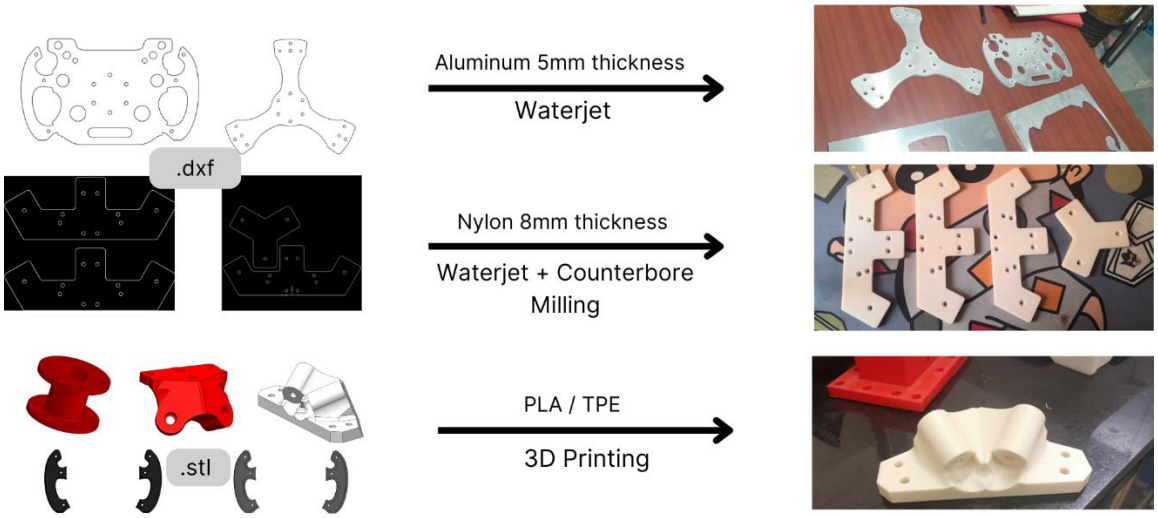


Figure 4.1 Various manufacturing process involved.

The base supports were cut in a similar manner with two nylon plates of dimensions 300x300mm and thickness 8mm. These 4 supports were placed in two .dxf files with two parts in each for cutting. These supports were to also act as counterbore, so after waterjet, we did milling of 10 mm diameter in the holes upto 6mm in depth so as to accommodate the M6 bolt head. All the wedges in the design were given as .stl files and were 3D printed using White PLA. The soft rubber cushioning of the joystick handle are to be printed in soft black TPE. Only the bushing are left, rest all the components were manufactured successfully.

4.3 Manufacturing of the base

4.31 Cutting and welding

In order to make the base frame successfully we need to apply various processes to get its manufacturing correctly. The processes followed in order to accurately manufacture the base frame is showcased clearly in Figure 4.2.

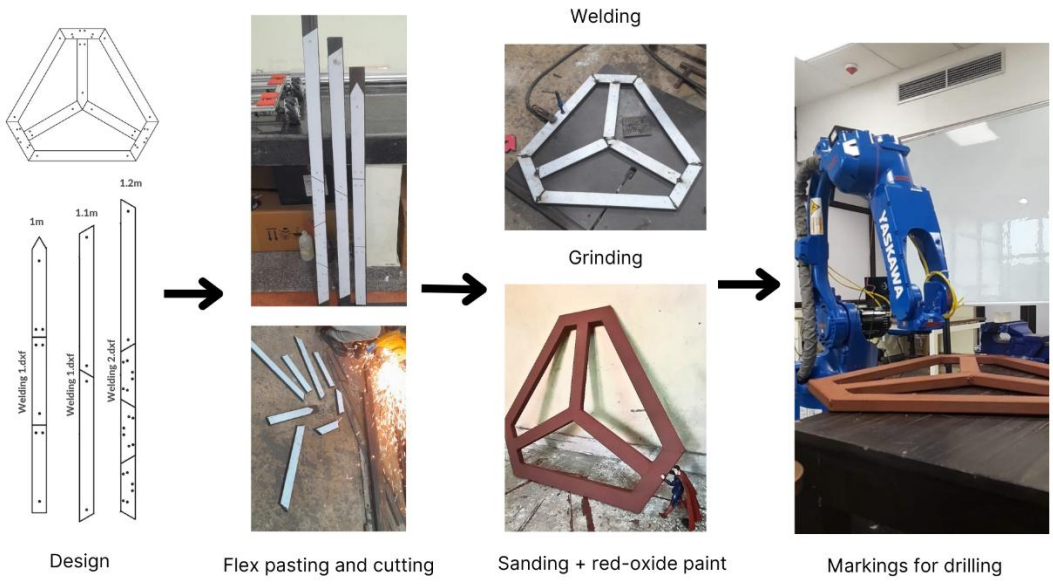


Figure 4.2 Processes involved in manufacturing of the base frame

The initial design of the base was split into three metal pipes of lengths little less than 1m, 1.1m and 1.2 m as shown in Fig 4.2. This laid the foundation for easy cutting of the pipes. We got the .dxf file of the layout printed on a plastic flex sticker with 1:1 size ratio, and pasted it on the steel pipes. With the layout in place, we easily cut all parts and proceed for welding the structure

4.32 Post processing and Drilling

After welding, the structure was not flat at its base and was quite uneven. So, we proceeded to grind the welded edges with a handle held angle grinder. Grinding was not done too deep, so as to avoid breaking of weld. After welding the entire frame was sanded and smoothened out. Priemer of red oxide was applied on the structure which will also help to bond any paint coating on it. Final coating of paint is left. As expected, there were errors in the frame and the dimension varied a little in final part. So rather than following the flex to cut out the holes, we proceeded to use robotic arm in order to mark the drilling point. We defined the origin and coordinate axis to one end point of the frame. The robotic arm was calibrated keeping that point as origin. The drilling coordinate point were taken from the center of holes in the CAD model, and fed into the teach

pendant. This gave us accurate locations of drilling centers. And also we were able to calculate how much error actually occurred because of the manufacturing processes.

4.4 Errors in manufacturing

Due to the cutting and welding of the base frame small human errors are unavoidable. These errors matter a lot when we are manufacturing a precise system as ours. Small changes can easily alter the symmetry and kinematics of our model, thus giving erroneous results. Apart from the dimension error in the base frame, there were other finishing errors in parts such as joystick handle, whose edges were quite sharp and needed filling. Also the base frame required good amount of grinding and sanding as mentioned before.

The dimensional error of the frame was avoided by not relying on initial drilling points, but rather locating them separately from the CAD file. Initial plan was to directly mill the holes using the CNC VMC mill but the limiting size of the machine was 500mmx500mm, while our frame dimension was 600m x 700mm. So, we proceeded to mark the points separately via feeding the points from CAD model to the teach pendant of the Yaskawa gp -12 robot. The robot was calibrated to one edge of the frame exactly as in CAD model. We found the error in the dimension of the frame to be 6mm less radius than expected. So rather than changing the drilling location, we will just audit this error in the final kinematic calculation of the robot.

5. ASSEMBLY AND COMPLETION OF THE ROBT

5.1 Building of the robot

After the Manufacturing of the robot, we finally move on to the assembly of the robot. We first coat the base of the stewart platform with white enamel coating. After the coating dries (It takes time to dry due to red oxide paint) we can easily assemble the nylon plates. This step can be a bit challenging due to paint blocking the holes and inclination in drilling in base. After assembling the nylon plates, we move on to attaching the base wedges. After that I move on to assembling the top portion. We attach the 3D printed handles after sanding to the steering handle. Then I attached the top wedges to the top plate. After that we connect the top plate and steering to with the connector and finish the assembly of the top and bottom portions of the robot.



Figure 5.1 Assembly of the Stewart Joystick

5.2 Final appearance of the robot

So, after completing the base and the top assembly, we can assemble the robot together to build the final product. We use 6 linear encoders, and attach it to the base and top wedges. We use 3D printed bushings as spacers to provide accurate distancing and motion. After screwing all the parts together, the bot is complete. There was minimal misalignment observed and the robot was quite sturdy. Overall the bot has 6 degrees of freedom and can be used as joystick for controlling various industrial parallel platforms in the industry.



Figure 5.2 Finished 6 DOF Stewart joystick

6. Conclusions and future scope

6.1 Conclusion

This project marked the successful completion of all major milestones in designing and building the 6-DoF Stewart Gough Platform for Potentiometer based Joystick, demonstrating a comprehensive understanding of its core aspects. From developing the CAD model to manufacturing the components, the work involved hands-on experience with advanced techniques such as waterjet cutting, milling, and 3D printing, alongside traditional methods like cutting, grinding, filing, and welding. Additionally, it provided valuable insights into material procurement and component arrangement, rounding out a practical learning experience. This project not only deepened technical skills but also reinforced the importance of meticulous planning and execution, paving the way for future advancements in robotics and mechanical design.



Figure 6.1 Final appearance of the robot

6.2 Future work

1) Solve the Kinematics specified to out design with error consideration
2) Build the electronic circuit for the robot
3) Learn and implement various control algorithms
4) Use Machine Learning to accurately control the robot

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